University of El Oued
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Master 2 – Artificial Intelligence and Data Science

Big Data Analytics

Practical Work TP04

Containerization and Big Data Environment Deployment Using Docker

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1 Introduction

Docker is a containerization platform that allows applications and their dependencies to be packaged into portable images and executed in isolated containers.

In this practical work, we explore the basic concepts of Docker, build an image containing a dataset, run containers, publish an image to Docker Hub, configure a Docker network, and use Docker Compose to simplify project deployment and management.

2 Objectives

The main objectives of this practical work are:

- Verify the Docker installation.
- Create a Docker Hub account.
- Build a Docker image containing the project dataset.
- Run a container and verify that the dataset is accessible.
- Push and pull the image using Docker Hub.
- Create multiple containers and configure a Docker network.
- Use Docker Compose to manage the project.
- Document the work and publish the project on GitHub.

3 Project Structure

The project is organized as follows:

```
TP04-Docker/
|-- Dockerfile
|-- compose.yaml
|-- data/
|   ‘-- train.csv
|-- README.md
‘-- screenshots/
```

The file `train.csv` is the dataset used in the project. The Dockerfile defines how the image is built, while the Compose file describes the service configuration.

4 Creating the Docker Image

4.1 Dockerfile Configuration

The following Dockerfile uses a lightweight Python image, sets the working directory, copies the dataset, and defines a command to verify the contents of the data directory.

```
FROM python:3.12-slim

WORKDIR /app

COPY data /app/data

RUN python -c "import os; print('Data files:', os.listdir('/app/data'))"

CMD ["python", "-c", "import os; print('TP04 Docker container started
    successfully'); print('Files:', os.listdir('/app/data'))"]
```

4.2 Explanation of Dockerfile Instructions

- `FROM`: Specifies the base image.
- `WORKDIR`: Sets the working directory inside the image.
- `COPY`: Copies the local dataset into the image.
- `RUN`: Executes a command during image building.
- `CMD`: Defines the default command executed when the container starts.

4.3 Building the Image

The image is built using the following command:

```
docker build -t tp04-analysis:v1 .
```

The option `-t` assigns a name and tag to the image. The dot indicates that the current directory is the build context.

4.4 Running the Container

The image is tested with:

```
docker run --rm tp04-analysis:v1
```

The expected output is:

```
TP04 Docker container started successfully
Files: ['train.csv']
```

This output confirms that the container executed successfully and that the dataset file is accessible inside it. The option `-rm` removes the container after it exits.

5 Publishing the Image on Docker Hub

Docker Hub is a registry that allows users to store, share, and download Docker images.

The following commands are used to publish and retrieve the project image:

```
docker login

docker tag tp04-analysis:v1 ghaya01/tp04-analysis:v1

docker push ghaya01/tp04-analysis:v1

docker pull ghaya01/tp04-analysis:v1
```

The commands perform the following operations:

- `docker login`: Authenticates the user.
- `docker tag`: Assigns the repository name and version tag.
- `docker push`: Uploads the image to Docker Hub.
- `docker pull`: Downloads the image from Docker Hub.

Docker Hub repository:
https://hub.docker.com/r/ghaya01/tp04-analysis

6 Creating a Docker Network

A Docker network allows containers to communicate with each other using network connectivity and container names.

First, the network is created:

```
docker network create tp04-network
```

Three containers are then created on the same network:

```
docker run -d --name tp04-container1 --network tp04-network -p 8081:80 ghaya01/
    tp04-analysis:v1

docker run -d --name tp04-container2 --network tp04-network -p 8082:80 ghaya01/
    tp04-analysis:v1

docker run -d --name tp04-container3 --network tp04-network -p 8083:80 ghaya01/
    tp04-analysis:v1
```

The main options are:

- `-d`: Runs the container in detached mode.
- `-name`: Assigns a name to the container.
- `-network`: Connects the container to the specified network.
- `-p`: Maps a host port to a container port.

The containers and network can be inspected using:

```
docker ps
docker network inspect tp04-network
```

Connectivity between containers was tested, and the network communication test was successful.

Port mapping does not automatically create a web server. An application must be listening on the mapped container port for HTTP requests to receive a response.

7 Using Docker Compose

Docker Compose simplifies the management of application services by defining their configuration in a single YAML file.

7.1 Compose Configuration

The `compose.yaml` file contains:

```
services:
  data-analysis:
    build: .
    container_name: tp04-analysis
    volumes:
      - ./data:/app/data:ro
    restart: "no"
```

Explanation:

- `services`: Defines the services in the project.
- `build: .`: Builds the image from the current directory.
- `container_name`: Sets the container name.
- `volumes`: Mounts the local data directory into the container.
- `ro`: Makes the mounted data read-only from inside the container.
- `restart: "no"`: Disables automatic restart after the container stops.

7.2 Running the Compose Project

The project is built and started using:

```
docker compose up --build
```

To inspect the containers:

```
docker compose ps -a
```

To stop and remove the Compose-managed containers:

```
docker compose down
```

The analysis container exits after completing its command. This is expected because the configured command is a short process rather than a continuously running service.

8 Discussion Questions and Answers

8.1 What is Docker and why is it useful in data science?

Docker is a containerization platform that packages an application with its dependencies into an image. It is useful in data science because it helps reproduce environments, share projects, and deploy applications consistently across different machines.

8.2 What is the difference between an image and a container?

A Docker image is a template containing the files and instructions required to run an application. A container is an instance created from that image in which a process can execute.

8.3 Explain FROM, WORKDIR, COPY, RUN, and CMD.

- `FROM` selects the base image.
- `WORKDIR` sets the working directory.
- `COPY` copies files into the image.
- `RUN` executes commands during the build.
- `CMD` specifies the default command at startup.

8.4 What problem does Docker Compose solve?

Docker Compose simplifies the configuration and management of applications containing one or more services. It stores their configuration in a YAML file and allows them to be started and stopped using simple commands.

8.5 Compare docker run and docker compose up.

The `docker run` command starts an individual container with options supplied directly in the command line.

The `docker compose up` command reads the Compose configuration and starts the services defined in that file.

8.6 How can Docker Compose be extended with a database?

A database service, such as MySQL or PostgreSQL, can be added to the Compose file. Its image, environment variables, and persistent storage can be configured. The analysis service can then communicate with the database through the Compose network. Real credentials should not be committed to a public repository.

9 Conclusion

This practical work introduced the fundamental concepts of Docker and containerization. We built and tested a Docker image containing the project dataset, published the image to Docker Hub, created containers connected to a Docker network, and used Docker Compose to simplify project management.

These activities demonstrate how containerization can improve environment consistency, portability, and the deployment of data science projects.